YEAR 7 MATHS · CHAPTER 7 · MIND MAP

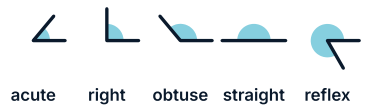
# Geometry

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## 1 Points, lines and angles

Acute under 90°, right 90°, obtuse 90° to 180°, straight 180°, reflex over 180°.

A full revolution is 360°

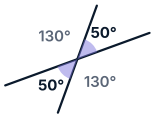



## 2 Adjacent and opposite angles

On a straight line, angles add to 180°. Vertically opposite angles are equal.

130° on a straight line: the other angle is **50°**

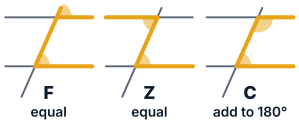
Complementary angles add to 90°




## 3 Parallel lines

Corresponding angles are equal (F), alternate angles are equal (Z), co-interior angles add to 180° (C).

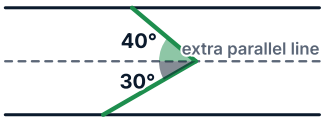
75°: co-interior angle **105°**




## 4 Compound problems

Draw an extra parallel line through the angle to split it.

Parts of 40° and 30°: the whole angle is **70°**



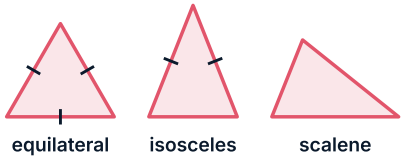

## 5 Triangles

By sides: equilateral, isosceles, scalene. By angles: acute, right, obtuse.

Equilateral: three equal sides

Isosceles: two equal sides

Scalene: no equal sides

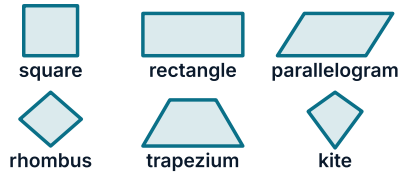



## 6 Quadrilaterals and polygons

Square, rectangle, parallelogram, rhombus, trapezium, kite.

Polygons: pentagon 5 sides, hexagon 6, octagon 8

Rhombus: four equal sides




## 7 Angle sum of a triangle

The angles of a triangle add to 180°.

50° and 60°: the third is **70°**

Isosceles with top angle 40°: each base angle is **70°**

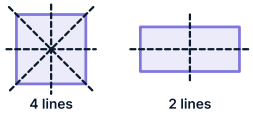



## 8 Symmetry

Line symmetry: it folds onto itself. Rotational symmetry: it fits onto itself as it turns.

Square: 4 lines, order 4

Rectangle: 2 lines, order 2




## 9 Reflection and rotation

Reflection: a mirror image. Rotation: a turn about a centre.

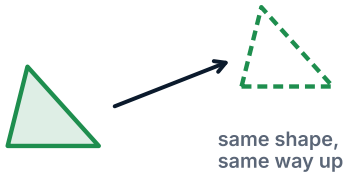
Give a rotation's centre, angle and direction, such as 90° clockwise




## 10 Translation

A slide, with no turning or flipping.

Right 4, up 1: $(2, 1) \rightarrow (6, 2)$

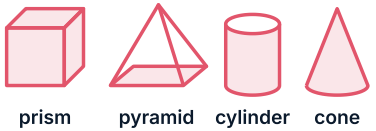



## 11 Drawing solids

Prisms, pyramids, cylinders, cones and spheres.

Draw top, front and side views

Isometric dot paper makes 3D drawing easier




## 12 Nets of solids

A net folds up into the solid.

Cube: 6 squares

Triangular prism: 2 triangles and 3 rectangles

Square pyramid: 1 square and 4 triangles

